ECE 329 **Homework 8** **Due: Thursday, October 22, 2026, 11:59 PM**

- Homeworks are due Thursday at 11:59:00 p.m. Late homework will not be accepted.
- Homeworks are to be turned in to Gradescope.
- Any deviation from the following steps will result in a 5 point penalty:
  - Write your name, netID, and section ONLY on the first page of your submission.
  - Start each new problem on a new page.
  - Scan the homework as a PDF (rather than taking pictures). Use a free scanning app if you do not have access to a photocopier.
  - Upload the PDF to Gradescope and tag each problem's location in the PDF.
- Each student must submit individual solutions for each homework. You may discuss homework problems with other students registered in the course, but you may not copy their solutions. If you use any source outside of class materials that we've provided, you must cite every source that you used.
- Use of homework solutions from past semesters is not allowed and is considered cheating. Copying homework solutions from another student or generative AI is considered cheating.
- Penalties for cheating on homework: a zero for the assignment on the first offense, and an F in the course on the second offense.

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Reading Assignment: Kudeki: Lectures 20-22

1. In a homogeneous lossess dielectric with $\epsilon = \epsilon_r \epsilon_o$ and $\mu = \mu_r \mu_o$ a plane TEM wave with the following components is observed:

$$\mathbf{E} = \hat{x} u(t - \frac{z}{c/2}) + \hat{y} g(t - \frac{z}{c/2}) \, \frac{\mathrm{V}}{\mathrm{m}}$$

and

$$\mathbf{H} = \hat{x}(\frac{40z}{c} - 20t) + \hat{y} \frac{1}{60\pi} u(t - \frac{2z}{c}) \, \frac{\mathrm{A}}{\mathrm{m}},$$

where $u(t)$ denotes the unit-step function and c is the speed of light in free space. Using the above information,

   a) Determine the intrinsic impedance η for the medium.

   b) Determine the propagation velocity v.

   c) Determine ϵ_r and μ_r.

   d) Function $g(t)$.

2. Consider a pulse of sheet current $\mathbf{J}_s(t) = \hat{y} \, 2\, t \;\mathrm{rect}(\frac{t}{\tau})$ A/m, where $\tau = 2\,\mu$s on the $z = 0$ plane embedded in free space and $rect(\frac{t}{\tau})$ is the unit rectangular function with width τ defined as:

$$rect(\frac{t}{\tau}) = \begin{cases} 0 & for\, |t| > \frac{\tau}{2} \\ 1 & for\, |t| < \frac{\tau}{2} \end{cases}$$

   The sheet will generate TEM waves which will propagate away from the current source in the $\pm\hat{z}$ directions.

   a) Write the expressions for $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$.

   b) Plot the field components for $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$ as a function of t at $z = 1500$ m.

   c) Repeat (b) for $z = -1500$ m.

   d) Plot the field components for $\mathbf{E}(z,t)$ and $\mathbf{H}(z,t)$ as a function of z for $t = 6\,\mu$s.

   e) Determine the TEM wave energy radiated per unit area (in J/m^2 units) by the current pulse. **Hint:** integrate the power injected per unit area, $-\mathbf{J}_s \cdot \mathbf{E}$, over the duration of pulse $\mathbf{J}_s(t)$.

3. Given a uniform TEM plane wave with $\mathbf{E}(z,t) = \cos(\omega t + \beta z) \, \hat{y} \, \frac{\mathrm{V}}{\mathrm{m}}$ propagating through a homogeneous perfect dielectric having $\epsilon = 3\epsilon_o$ and phase speed $v_p = \frac{2}{3}c$ and frequency $f = 300$MHz, determine the following in appropriate units:

   a) the relative permeability μ_r and the intrinsic impedance η of the medium

   b) the wavelength λ of the wave

   c) the phase constant β

   d) the corresponding $\mathbf{H}$ wavefield

   e) the instantaneous Poynting vector at $t = 10$ns on the $z = \frac{1}{2}$ m plane

   f) the instantaneous Poynting vector at $t = 10$ns on the $z = 1$ m plane

   g) in view of energy conservation and the results of parts (e) and (f), determine if energy is being stored into or released from the EM field at $t = 10$ns, within a cube of length $\frac{1}{2}$m located between the $z = \frac{1}{2}$ m and the $z = 1$ m planes.

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4. For each of the following plane TEM waves in free space:

   a) $\mathbf{E}_1 = 10\cos(\omega t - \beta x)\hat{y}$ V/m

   b) $\mathbf{H}_2 = 5\cos(\omega t + \beta z + \frac{\pi}{3})\hat{x} + 5\sin(\omega t + \beta z - \frac{\pi}{6})\hat{y}$ A/m

   c) $\mathbf{H}_3 = \sin(\omega t - \beta z + \frac{\pi}{4})\hat{x} - 2\sin(\omega t - \beta z - \frac{3\pi}{4})\hat{y}$ A/m

   d) $\mathbf{E}_4 = 2\cos(\omega t + \beta y - \frac{\pi}{2})\hat{x} - 2\sin(\omega t + \beta y)\hat{z}$ V/m

   e) $\mathbf{H_5} = \cos(\omega t - \beta y)\hat{x} + \sin(\omega t - \beta y - \frac{\pi}{4})\hat{z}$

      i. Determine the expression for **H** or **E** that accompanies the given wave field

      ii. Find the corresponding field phasors $\tilde{\mathbf{E}}$ **and** $\tilde{\mathbf{H}}$ for all wave fields

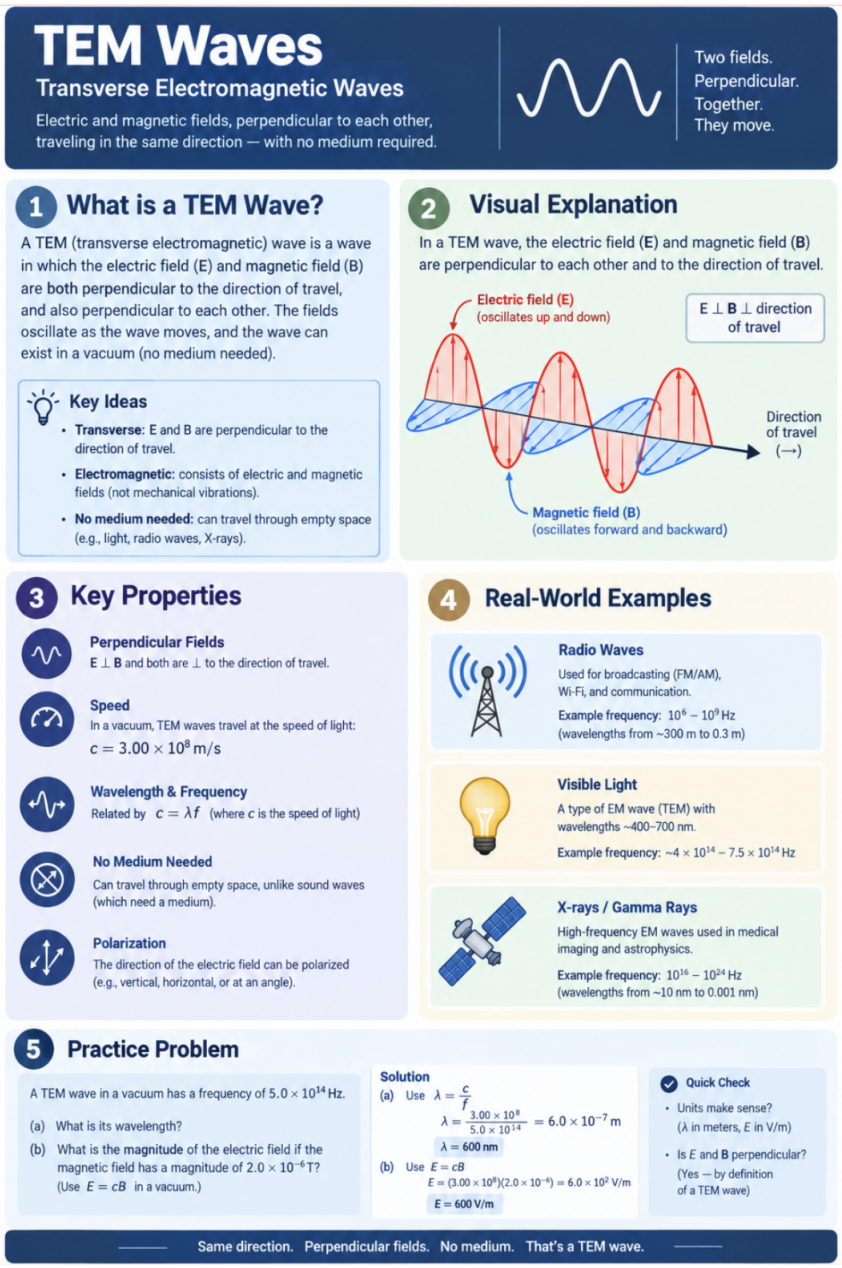
      iii. Find the expression for the instantaneous power that crosses a 1 m^2 area in the direction of the wave propagation

      iv. Find the time-averaged power that crosses a 1 m^2 area in the direction of the wave propagation. Hint: you should be able to answer this question based on either part (ii) or (iii)

5. Critical Thinking with Generative AI problem:
   Topic: **Wave propagation in free space**

   a) **Goal**
   Use generative AI tools to create at least two beginner-friendly, comprehensive infographics explaining the assigned concept. Your infographics should include **examples**, not just descriptions. Then critically evaluate what the AI gets right, wrong, or leaves unclear. The goal is not to produce a perfect infographic. The goal is to detect problems and explain your reasoning using course knowledge and evidence.
   Example infographic by ChatGPT, GPT 5.6 Luna:

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b) **Generate Infographics**

Generate at least two versions of a beginner-friendly infographic explaining the assigned topic. Each infographic should include the major ideas or relationships, appropriate equations, diagrams, labels, and at least one example.

You may generate the two versions using two different AI tools, two substantially different prompts, or a combination of both. You are encouraged to use follow-up prompts to clarify, correct, or improve the generated infographic.

Report:

i. AI platform(s) and model/version(s) used

ii. Your prompts and follow-up prompts (screenshots or text)

iii. Your chat history (screenshots or saved chat links)

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iv. Your two generated infographics.

c) **Choose and Evaluate One Infographic**

i. Identify one part of the infographic that is technically correct. Explain why it is correct using ECE 329 concepts. For example, the equation $E = cB$ seems incorrect because we learned $E = \eta_o H$ in class. However, since $\eta_0 = \sqrt{\mu_0/\epsilon_0}$, $c = 1/\sqrt{\mu_0\epsilon_0}$, and $B = \mu_0 H$, the equation $E = cB$ is also correct.

ii. Identify at least one error, ambiguity, misleading statement, or important omission. Explain why it is a problem and why it matters technically.

iii. Explain how you verified your conclusion. You may use course knowledge and, if helpful, credible external sources such as textbooks, academic resources, or technical articles. Cite any external sources you use.

iv. Explain how the infographic could be corrected or improved.

d) **Trust and Accessibility**

Identify one part of the infographic that you would trust and one part that you would verify before using. Explain why.

Briefly comment on whether the infographic is understandable for a student learning the topic for the first time. Consider organization, labels, notation, symbols, font size, color contrast, and whether the visuals could create confusion.

e) **Reflection**

Briefly reflect on your interaction with the AI:

i. How did your prompt(s) influence the AI's output?

ii. What limitations did you notice in the AI-generated content?

iii. How did follow-up prompting affect, or how could it have improved, the result?

Remember, the goal is not to produce a perfect AI infographic. The goal is to recognize and explain what the AI gets right, wrong, or leaves out.

f) **Important**

- Do not upload course-provided materials (lecture slides, notes, assignments, etc.) to the AI tool. If you would like to provide the AI with additional context, you may find publicly available materials related to the topic online and use or upload those instead.
- Do not rely only on a shared chat link. Links may expire or become inaccessible; save screenshots or another permanent copy of your interaction.
- Do not simply accept the AI's output as correct. Use your course knowledge and evidence to evaluate it.
- Do not include private or personally identifiable information in your prompts.

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